IEEE Access
Multidisciplinary : Rapid Review : Open Access Journal

Received 22 October 2024, accepted 16 December 2024, date of publication 25 December 2024, date of current version 2 January 2025.

Digital Object Identifier 10.1109/ACCESS.2024.3522285

RESEARCH ARTICLE

Tokenizers for African Languages

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This work was supported by the Ministry of Education of the Republic of Korea and the National Research Foundation of Korea (NRF-2022S1A5A2A03052880). This work was supported by Institute of Information & communications Technology Planning & Evaluation (IITP) grant funded by the Korea Government (MSIT) (RS-2022-II220043, Adaptive Personality for Intelligent Agents).

ABSTRACT Despite incredible development in the field of natural language processing (NLP), there has been a huge gap in the performance of NLP tasks between high-resource languages (HRLs) and low-resource languages (LRLs). African languages belong mainly to the LRLs, and one of the major contributing factors to the performance gap is tokenization, which plays a crucial role in NLP performance in general. Many recent studies on African languages often rely on multilingual tokenizers or general-purpose tokenizers, which are not optimized for African languages. This may lead to suboptimal performance in downstream NLP tasks. In this paper, we systematically analyze the performance of language-specific tokenizers for three African languages: Swahili, Hausa, and Yoruba. By experimental results on two classification tasks (i.e. sentiment classification and news classification), we found that the language-specific tokenizers for African languages consistently outperformed other monolingual tokenizers, with performance gaps of up to 5.43% in sentiment classification and 4.58% in news classification. We also found that multilingual tokenizers generally work well if they are trained in many African languages rather than global HRLs. For instance, African multilingual tokenizers outperformed global multilingual tokenizers by an average of 1.70% in sentiment classification and 1.41% in news classification. The largest observed improvement was 2.61% in news classification using Logistic Regression (LR). Based on the results, we suggest a method for choosing tokenizers when analyzing data or developing models for African languages.

INDEX TERMS Tokenizer, classification, African language, Swahili, Yoruba, Hausa.

I. INTRODUCTION

Recently, multilingual language models (mLMs) have made significant progress in bridging the linguistic divide by accommodating low-resource languages (LRLs) [1], [2], [3], [4]. Despite these advancements, their performance on LRLs remains suboptimal. One of the primary factors contributing to this disparity is tokenization. Different tokenizers produce different tokens for the same input text, making tokenization fundamental to the performance of language models (LM) [5], [6]. We argue that to improve the performance of LM in LRL, it is essential to develop tokenizers specifically tailored to the unique linguistic characteristics of these languages [7], [8].

Language-specific tokenizers capture the nuances and intricacies of LRLs better, leading to more accurate and effective language processing. To demonstrate the importance of language-specific tokenizers for LRLs, [9] highlighted the complexities involved in tokenization of the Kurdish language, due to the lack of standardized orthography and the extensive use of compound words. They further underscored the importance of tokenizers specifically designed to handle the linguistic nuances of LRLs, ensuring accurate segmentation and better performance in downstream natural language processing (NLP) tasks. Furthermore, [10] showed the importance of tokenization of subwords in addressing various challenges of LRLs using the Marathi language

The associate editor coordinating the review of this manuscript and approving it for publication was P. K. Gupta.

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in downstream NLP tasks. These studies demonstrate that language-specific tokenizers enhance the quality of text processing and advance NLP applications for these underrepresented languages.

African languages are among the LRLs where tokenization poses a predominant challenge. Languages such as Swahili, Hausa, and Yoruba have unique characteristics that challenge general-purpose tokenizers. For example, Swahili's agglutinative nature, which forms words by combining morphemes, presents performance limitations in tokenization [11]. Hausa's nominalization and polysynthetic system requires sensitivity to pitch variations [11], while Yoruba's reliance on tone and complex vowel harmony introduces additional difficulties for standard subword and multilingual tokenizers [12]. These features highlight the need for specialized tokenizers to handle the morphological richness and phonological rules of African languages.

Despite considerable efforts to develop language-specific tokenizers for LRLs, African languages have often been overlooked. Research has been conducted on creating multilingual tokenizers [1] and language-specific tokenizers [13], [14]. However, the most widely spoken African languages (e.g., Swahili, Hausa, and Yoruba) [15] lack sufficient improvements in tokenization, which impacts the performance of LMs in various NLP tasks [16]. Although there has been significant progress in NLP for African languages, such as sentiment analysis [17], African datasets [18], and African-specific LM [19], the foundational aspect of tokenization has not yet received the same level of attention. This limitation highlights the need for dedicated improvement and development of tokenizers specialized in the unique linguistic characteristics of African languages.

In this paper, we present a novel tokenizer specifically designed for African languages, including Swahili, Hausa, and Yoruba, addressing the unique linguistic challenges these languages pose. Our contributions are significant in several key aspects.

First, we demonstrate that our monolingual tokenizer outperforms well-known multilingual tokenizers by effectively capturing the morphological and phonological nuances of African languages, leading to more accurate word segmentation and representation. Second, we show that African multilingual tokenizers outperform general multilingual tokenizers trained not only in African languages but also on high-resource languages (HRLs), highlighting the importance of careful language selection when creating multilingual tokenizers. In all experiments, we compare various monolingual and multilingual tokenizers in sentiment classification (SC) and news classification (NC) tasks.

Following the introduction and detailed contributions, the remainder of this paper is structured as follows. In the related work section, we review existing research on tokenization strategies for LRLs and highlight the limitations addressed by our proposed approach. In the methodology section, we describe the design and implementation of our custom tokenizers for African languages, emphasizing how they handle the unique linguistic features of Swahili, Hausa and Yoruba. The experiment section details our experimental setup, including datasets, evaluation metrics, and comparison results, where we highlight the better performance of language-specific tokenizers for these languages. Finally, the conclusion summarizes our findings and offers guidance on selecting tokenizers for African languages.

II. RELATED WORK

Several studies have proposed or utilized tokenizers specifically for various natural language processing tasks. In the early stages, statistical methods such as n-gram models and word frequency were widely adopted, forming the foundation for tokenization. Reference [20] discussed these statistical methods and highlighted their effectiveness during the initial phases of language processing. However, these methods were soon recognized for their limitations, particularly in handling morphological complexity and syntactic variations. Likewise, [21] underscored the challenges faced by statistical models in managing complex linguistic structures, reinforcing the need for more advanced models in language processing.

A key driver for the development of advanced tokenization methods was the introduction of subword tokenization techniques, such as Byte Pair Encoding (BPE) [22] and WordPiece tokenization [23], later popularized by [1]. These subword tokenization methods struck a balance between word-level and character-level tokenization, offering a reliable approach for handling out-of-vocabulary words and enhancing the efficiency of language models. However, addressing language-specific requirements remains a significant challenge, often resulting in suboptimal performance for low-resource languages.

In efforts to address the specific challenges faced by low-resource languages, recent research has sought to enhance subword tokenizers and develop more language-agnostic approaches. Reference [24] introduced a language-agnostic tokenization method that processes raw text without the need for pre-tokenization, simplifying the pipeline for low-resource languages. Additionally, later works, such as [25], further refined these techniques by introducing gradient-based approaches to improve tokenizer efficiency and adaptability across diverse languages and linguistic complexities. While these approaches handle contiguous sequences of characters effectively, they struggle with non-concatenative morphology, where morphemes are discontinuous. Additionally, gradient-based approaches do not provide a comprehensive solution for tokenizing low-resource languages.

Instead of using language-specific tokenizers for low-resource languages, some studies have employed multilingual tokenizers designed to cover a broader range of languages. Reference [26] demonstrated that transfer learning and multilingual models significantly improved the performance of NLP tasks in low-resource settings by leveraging cross-lingual embeddings. Similarly, [27] explored both

the benefits and limitations of multilingual models. Their findings suggest that while multilingual models benefit from positive transfer between related languages, they also suffer more from negative transfer compared to auto-regressive models. This issue arises from the models' struggles to generate valid tokens with the desired output length and their frequent generation of invalid words, a problem that is magnified in multilingual scenarios. Furthermore, [28] highlighted the disparities in tokenization from tokenizers trained on multilingual data, which disadvantage certain language communities. In essence, multilingual tokenizers fail to generalize effectively across all languages.

In another study, [13] introduced Word-Expression-Based (WEB) tokenization for Fon, a low-resource African language. This research highlights the limitations of standard tokenization techniques and demonstrates that WEB tokenization, which incorporates human-guided super-word tokenization, more effectively captures the linguistic nuances of the Fon language. WEB tokenization shows significant improvements in neural machine translation tasks between Fon and French, underscoring the importance of developing language-specific tokenizers for underrepresented languages. However, it requires substantial human involvement to create the superword tokenization, a process that is both time-consuming and labor-intensive. This reliance on human input limits the scalability and practicality of the method for other low-resource languages that lack sufficient linguistic resources.

In this paper, we address the often overlooked tokenization challenges of African languages, including Swahili, Hausa, and Yoruba. We show that our African-specific tokenizers significantly outperform general-purpose and multilingual tokenizers by effectively capturing the unique morphological and phonological features of these languages. Compared to existing solutions, our approach excels in managing the linguistic complexities of African languages, resulting in improved word segmentation and representation. Extensive experiments show that our tokenizer not only enhances text processing quality but also significantly boosts the performance of downstream NLP tasks, such as sentiment analysis and news classification. This study underscores the critical need for tailored tokenization strategies to improve language model performance for low-resource languages.

III. MATERIALS AND METHODS

A. DATASET

For each of our target languages (e.g. Swahili, Hausa, and Yoruba), we obtained two datasets "serving different purposes: (1) a larger, unlabeled training dataset for tokenizer pretraining, D^L_{tok} and (2) a smaller, labeled downstream task-specific dataset, D^L_{task}, for sentiment classification (SC) and news classification (NC). The larger dataset is used to train the tokenizer by leveraging the language's raw text, while the smaller, labeled dataset is utilized for evaluating downstream tasks, as it contains sentiment and news labels. The task-specific datasets are smaller due to the effort required for labeling, ensuring that we evaluate the tokenizer on relevant, real-world tasks. The sources of the datasets are summarized as follows.

- $D^{Swahili}_{tok}$: a pre-training dataset of [19].
- $D^{Swahili}_{SC}$: a sentiment classification dataset of [19].
- $D^{Swahili}_{NC}$: an online Tanzanian news website 'Mwananchi'.
- D^{Hausa}_{tok} and D^{Yoruba}_{tok}: unlabeled texts from [18].
- D^{Hausa}_{NC}, and D^{Yoruba}_{NC}: the Masakha news collection from [18].
- D^{Hausa}_{SC}, D^{Yoruba}_{SC}: the Naijasenti dataset from [29].

The raw texts were cleaned by removing punctuation, extra spaces, and converting all text to lowercase. Table 1 summarizes the dataset statistics, where each sample in $D^L tok$ is a single sentence, and each sample in $D^L task$ consists of one or more sentences.

TABLE 1. The number of instances in different datasets for Swahili, Hausa, and Yoruba languages. The larger dataset is used for tokenizer pretraining (unlabeled), while the smaller datasets are used for downstream tasks, including sentiment classification (SC) and news classification (NC), which are labeled.

Language	Tokenizer training (D^L_{tok})	SC (D^L_{SC})		NC (D^L_{NC})	
		training	test	training	test
Swahili	688,867	4,284	1,066	1,516	379
Hausa	805,787	13,479	3,370	2,028	508
Yoruba	562,238	8,489	2,123	1,311	328

B. TRAINING TOKENIZERS

We selected the WordPiece tokenization algorithm for its ability to generate meaningful subword units, which is crucial for languages with complex morphology, such as Swahili, Hausa, and Yoruba. The initial vocabulary consisted of individual characters from each language corpus, and it was enriched with frequent character sequences and common words. The WordPiece algorithm iteratively merges the most frequent character or subword tokens in the corpus until the desired vocabulary size is reached. This approach balances the capture of meaningful subword units with maintaining a manageable vocabulary size. It is designed to effectively handle common morphological constructs specific to each language, such as affixes in Swahili, nominalization in Hausa, and vowel harmony in Yoruba.

Figure 1 illustrates the model in two parts: (1) Tokenizer training, where the three language-specific tokenizers are created, and (2) Tasks section, where we perform downstream tasks, including sentiment classification and news classification. We explain each part of the figure as follows:

- D_{swa}, D_{hau}, D_{yor}: large unlabeled text datasets for Swahili, Hausa, and Yoruba used for tokenizer training.
- Custom tokenizer: Processes raw text into tokenized sequences composed of subwords or tokens.
- Tokenized dataset: Output from the tokenizer in a JSON format, used for downstream tasks.

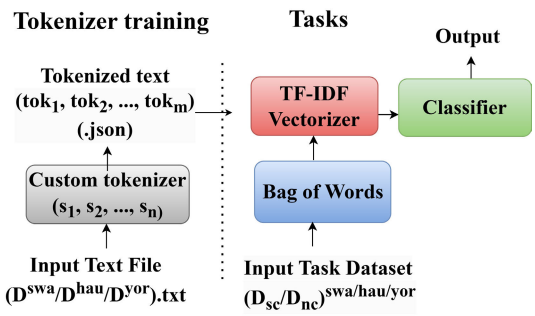

FIGURE 1. An overview of the tokenizer training and evaluation process for Swahili, Hausa, and Yoruba. The pipeline involves pretraining tokenizers on large, unlabeled datasets, followed by tokenization and feature extraction. The tokenized sequences are then evaluated on downstream tasks, including sentiment classification (SC) and news classification (NC), using traditional machine learning models such as Support Vector Machines (SVM), Random Forests (RF), and Linear Regression (LR).

- D_{SC}, D_{NC}: labeled datasets for sentiment classification (SC) and news classification (NC).
- Bag-of-Words (BoW): transforms the tokenized text into feature vectors based on token occurrence.
- TF-IDF vectorizer: processes the BoW representation to capture the relative importance of tokens in the corpus.
- SVM, RF, LR: machine learning classifiers trained on TF-IDF vectors for SC and NC tasks, providing predicted class labels as output.

C. EVALUATION TASKS

Once the tokenizers are trained, we evaluate them on two tasks: sentiment classification (SC) and news classification (NC). The SC task focuses on identifying the polarity of text, while the NC task requires distinguishing broader contextual differences between topics. This distinction allows us to assess how well the tokenizers generalize across different downstream tasks. The $D^{Swahili}NC$ dataset contains five classes: national, international, politics, sports, and business. The $D^{Yoruba}NC$ dataset includes five classes: entertainment, politics, health, sports, and religion. The D_{NC}^{Hausa} dataset has seven classes: business, religion, technology, politics, entertainment, sports, and health. For sentiment classification, the $D^{Swahili}SC$ dataset includes three classes: positive, negative, and neutral. The $D^{Yoruba}SC$ dataset has three classes: positive, negative, and ambiguous. Similarly, the D_{SC}^{Hausa} dataset contains three classes: positive, negative, and ambiguous.

Language models (LM) such as Bidirectional Encoder Representations from Transformers (BERT) [1] and the Generative Pre-trained Transformer (GPT) series [30] have demonstrated superior performance in many NLP tasks. However, we do not employ these models in our study. This decision allows us to focus on evaluating the effectiveness of the tokenizers without leveraging the advantages provided by LMs. Instead, we adopt conventional machine learning models such as Support Vector Machines (SVM) [31], [32], Logistic regression (LR), and Random Forests (RF) [33].

Given an arbitrary raw text, it is first converted into a sequence of tokens, as shown in Table 2. The token sequence is then transformed into a Bag-of-Words (BOW) representation, which is a V-dimensional integer vector where V represents the vocabulary size. We then apply TF-IDF to the vector. The classifiers use the resulting TF-IDF vectors as feature vectors to generate predictions.

TABLE 2. Example of the text conversion process into feature vectors given a raw Swahili sentence "Ninapenda kusoma vitabu" ("I love reading books"). The conversion process contains tokenization, BoW representation, and Term Frequency-Inverse Document Frequency (TF-IDF) vector.

Process	Example
Raw text	Ninapenda kusoma vitabu
Token sequence	[Ninapenda, kusoma, vitabu]
BoW representation	[1, 1, 1, 0, 0, 0]
TF-IDF vector	[0.5, 0.5, 0.5, 0, 0, 0]

In this study, we employed three traditional machine learning algorithms to evaluate the performance of the tokenizers on downstream tasks, specifically sentiment classification (SC) and news classification (NC):

- Support Vector Machine (SVM): A supervised learning algorithm used for classification tasks that identifies the optimal hyperplane to separate data points into different classes in high-dimensional space. SVM is particularly effective in high-dimensional feature spaces, such as the TF-IDF vectors used in our experiments, and was chosen for its robustness in handling sparse data representations.
- Random Forest (RF): An ensemble learning method that combines multiple decision trees to improve classification. Each tree is trained on a random subset of the data, and the final prediction is determined by the majority vote of the trees. RF is well-suited for handling complex, non-linear relationships, making it an ideal choice for capturing the linguistic diversity represented by our tokenizers.
- Logistic Regression (LR): A linear model used for binary and multi-class classification, which estimates the probability of a data point belonging to a particular class using a logistic function. LR is known for its simplicity and interpretability, and was included to compare its performance against more complex models like SVM and RF.

In this study, we opted for traditional machine learning models, including Support Vector Machines (SVM), Logistic Regression (LR), and Random Forest (RF), for several key reasons. These models are highly interpretable, allowing us to easily understand how tokenized input affects classification performance, which is crucial when evaluating different tokenization methods. Moreover, they are well-suited for small datasets, such as those used in our sentiment

classification (SC) and news classification (NC) tasks. Traditional models like SVM, LR, and RF perform effectively with feature engineering techniques such as TF-IDF, whereas deep learning models may overfit or require significantly larger datasets to achieve similar results. Additionally, these models are widely recognized as baselines in text classification tasks, offering a strong reference point for assessing tokenizer quality.

The pre-trained tokenizers (e.g., $\mathbf{T_{Serengeti}}$, $\mathbf{T_{AfriBERTa}}$) are used to tokenize the input text, creating token sequences that are converted into feature vectors (e.g., TF-IDF vectors). These feature vectors are then used as input for the traditional machine learning models, such as Support Vector Machines (SVM), Random Forest (RF), and Logistic Regression (LR). While the tokenizers provide consistent tokenized text across experiments, the machine learning models may perform differently based on their ability to process the tokenized input. Each model uses a distinct algorithm for classification, which leads to variations in performance for the same language depending on the model used.

IV. EXPERIMENTS

For both tasks (i.e., sentiment classification and news classification), we employed three traditional machine learning models: Support Vector Machine (SVM), Random Forest (RF), and Logistic Regression (LR). We utilized "'scikit-learn' tool'', and followed the default hyper-parameter settings. The models are trained on the training set, and evaluated on the test set, where the statistics of the task datasets are described in Table 1. We used accuracy as the evaluation metric. All accuracies are averages of five independent runs and are accompanied by standard deviations.

A. MONOLINGUAL TOKENIZERS

To examine the impact of language-specific tokenizer for our target languages (Swahili, Hausa, and Yoruba), we compared them with tokenizers of other high-resourced languages (HRL) such as English, Spanish, Arabic, and French. The four high-resourced languages were selected in line with the studies by [34] that suggested that LRLs benefit from HRLs of the similar script in the multilingual settings. For example, the primary difference between the Swahili and English scripts is the absence of the characters 'x' and 'q' in Swahili. Forty percent of Swahili words are influenced by the Arabic language. Hausa traditionally is written in Arabic script with increasing mix of Latin script in modern texts. Yoruba is also primarily written in the Latin alphabet which is used for English, Spanish, and French languages.

We selected the four HRLs tokenizers, which were trained on rich datasets, in line with the notion that low-resource languages (LRLs) benefit from high-resource languages in multilingual settings [1], and that LRLs tend to benefit more from languages with similar script [35]. This makes these tokenizers a suitable comparison for our LRL-specific tokenizers.

The differences in performance for the same language across SVM, RF, and LR can be attributed to the unique algorithms used by each model. While the tokenized input remains consistent, SVM is better suited for handling high-dimensional feature spaces, RF excels in capturing nonlinear patterns, and LR provides robust linear classification. These variations in model behavior lead to the observed differences in results.

The sentiment classification results for Swahili, Hausa, and Yoruba using different tokenizers are summarized in Table 3. For a word-based tokenizer $T_{Default}$, the vocabulary size is dynamically determined according to the given dataset. For Swahili, Hausa, and Yoruba, the vocabulary size of $T_{Default}$ is 10k, 22k, and 16k respectively. For other tokenizers, the vocabulary size is 30k. Table 3 important findings.

Interestingly, the African tokenizers ($T_{Swahili}$, T_{Yoruba}, and T_{Hausa}) performed well not only for their target languages but also across other African languages, suggesting shared linguistic features between the three LRLs, such as morphological complexity or shared phonological rules.

Even though the HRL tokenizers were trained on larger datasets, they generally performed poorly on the three African languages. Exceptionally, for the Yoruba language, $T_{English}$ gave comparable performance to the African tokenizers; this makes sense as Yoruba is mostly written in Latin alphabet. In contrast, the tokenizers of three African languages work pretty well for each other languages; for instance, T_{Hausa} give comparable performance to $T_{Swahili}$ and T_{Yoruba} for their languages. This may indicate that the three African languages have linguistic bonds. This is inline with an observation by [35] that, abstract linguistic similarities, rather than surface-level features like vocabulary overlap, drive the benefits of multilinguality. This is evident in our experiments, where the performance of African tokenizers showed greater consistency across LRLs than with HRLs. For instance, T_{Yoruba} outperformed the Arabic tokenizer on Hausa datasets, indicating stronger transference between LRLs. Similarly, Swahili performed better with the three African tokenizers than with the English tokenizer, despite the script similarity between Swahili and English. Second, for African languages, language-specific tokenizers exhibited the best performance. For examples, Swahili and Hausa tokenizers ($T_{Swahili}$ and T_{Hausa}) achieved the best accuracy for their target languages. Interestingly, for Yoruba, the word-based tokenizer $T_{Default}$ showed better performance than T_{Yoruba}. The main reason should be that the vocabulary of $T_Default$ is obtained from the task training data. In other words, its vocabulary is task-dependent, making it impractical in real-world scenarios. Other than $T_{Default}$, the language-specific tokenizer T_{Yoruba} is the best for Yoruba. This indicates that language-specific tokenizers play a crucial role in task performance for the three African languages.

The results in Table 4 show similar trends to Table 3; however, it is important to note that sentiment classification (SC) and news classification (NC) are distinct tasks with

TABLE 3. Sentiment classification accuracy for Swahili, Hausa, and Yoruba in comparison to monolingual tokenizers of high-resource languages (HRLs). The models include support vector machine (SVM), random forest (RF), and linear regression (LR) models.

Language	Models	$T_{Default}$	$T_{Swahili}$	T_{Hausa}	T_{Yoruba}	$T_{English}$	$T_{Spanish}$	T_{Arabic}	T_{French}
Swahili	SVM	**87.22±0.00**	85.53±0.00	82.75±0.01	82.82±0.00	82.41±0.00	80.65±0.00	76.49±0.00	80.63±0.00
	RF	90.13±0.01	**91.36±0.01**	90.98±0.01	90.50±0.01	91.04±0.01	90.83±0.01	90.38±0.01	90.85±0.01
	LR	89.32±0.00	**89.83±0.01**	82.63±0.01	81.83±0.01	79.92±0.01	79.95±0.01	70.42±0.01	77.19±0.01
	Avarage	88.89±0.00	**88.91±0.01**	85.45±0.01	85.05±0.01	84.46±0.01	83.81±0.01	79.10±0.00	82.89±0.00
Hausa	SVM	69.77±0.01	70.37±0.01	69.99±0.01	**70.56±0.01**	69.87±0.01	69.22±0.01	67.37±0.01	68.65±0.01
	RF	70.68±0.01	72.21±0.01	72.31±0.01	**72.40±0.01**	71.42±0.01	70.55±0.01	68.53±0.01	70.01±0.01
	LR	72.72±0.01	72.39±0.01	**74.10±0.01**	72.82±0.01	71.56±0.01	70.65±0.00	68.54±0.01	70.48±0.01
	Avarage	71.06±0.01	71.66±0.01	**72.13±0.01**	71.93±0.01	70.95±0.01	70.14±0.00	68.15±0.01	69.71±0.01
Yoruba	SVM	**67.66±0.00**	63.38±0.00	63.57±0.00	64.96±0.00	63.18±0.00	59.31±0.01	61.93±0.01	59.91±0.01
	RF	**65.08±0.01**	63.30±0.01	63.27±0.01	63.73±0.01	62.67±0.00	62.50±0.01	61.38±0.01	62.42±0.01
	LR	**69.17±0.01**	64.97±0.01	64.67±0.01	68.78±0.00	64.89±0.00	63.91±0.01	61.56±0.01	63.60±0.00
	Avarage	**67.30±0.00**	63.88±0.00	63.84±0.00	65.82±0.00	63.58±0.00	61.91±0.01	61.62±0.01	61.98±0.00

TABLE 4. News classification accuracy for Swahili, Hausa, and Yoruba in comparison to monolingual tokenizers of high-resource languages (HRLs). The models include support vector machine (SVM), random forest (RF), and linear regression (LR) models.

Language	Models	$T_{Default}$	$T_{Swahili}$	T_{Hausa}	T_{Yoruba}	$T_{English}$	$T_{Spanish}$	T_{Arabic}	T_{French}
Swahili	SVM	54.51±0.02	**57.20±0.01**	53.14±0.02	53.40±0.01	51.03±0.01	52.40±0.02	49.45±0.01	49.97±0.01
	RF	54.46±0.01	**56.25±0.01**	53.98±0.02	53.72±0.01	52.56±0.01	54.78±0.01	50.71±0.01	51.40±0.01
	LR	**59.53±0.01**	59.00±0.01	54.67±0.02	53.14±0.01	53.98±0.01	53.14±0.01	47.18±0.02	52.08±0.02
	Avarage	56.17±0.02	**57.48±0.01**	53.93±0.01	53.42±0.01	53.42±0.01	53.44±0.01	49.11±0.01	51.15±0.01
Hausa	SVM	70.35±0.04	**78.11±0.02**	75.12±0.03	77.01±0.03	71.57±0.02	68.94±0.03	67.24±0.03	69.49±0.03
	RF	83.35±0.02	82.24±0.01	**83.86±0.01**	82.09±0.01	82.48±0.01	81.85±0.01	79.72±0.01	81.46±0.02
	LR	83.43±0.01	82.91±0.01	**83.58±0.01**	82.44±0.01	82.64±0.01	82.32±0.01	79.25±0.01	82.05±0.01
	Avarage	79.04±0.02	**81.09±0.01**	80.85±0.01	80.51±0.01	78.90±0.01	77.70±0.02	75.40±0.02	77.33±0.02
Yoruba	SVM	**78.96±0.02**	73.72±0.01	74.09±0.01	76.95±0.02	64.70±0.02	51.89±0.02	60.18±0.02	47.80±0.02
	RF	**85.49±0.01**	81.16±0.01	82.32±0.01	85.24±0.02	81.04±0.03	82.62±0.02	78.35±0.02	81.10±0.02
	LR	**87.38±0.02**	82.32±0.01	83.05±0.01	86.34±0.02	84.09±0.01	84.33±0.02	81.52±0.02	83.17±0.01
	Avarage	**83.94±0.02**	79.07±0.01	79.82±0.01	82.84±0.01	76.61±0.02	72.95±0.01	72.95±0.01	70.69±0.02

different linguistic demands. SC focuses on identifying sentiment-related features such as polarity, while NC requires recognizing topic-based distinctions. Evaluating the tokenizers on both tasks helps ensure they perform well across a variety of NLP tasks, reflecting their robustness and generalization capabilities. While the results appear similar, the separation of tasks provides valuable insights into how the tokenizers handle different types of language processing challenges.

This again implies that the three African languages have linguistic bonds among them, and that language-specific tokenizers play a crucial role in task performance for these languages.. Interestingly, for Hausa language, $T_{Swahili}$ even showed better performance than T_{Hausa}, suggesting a strong linguistic bond between the two languages.

B. MULTILINGUAL TOKENIZERS

We further investigate several multilingual tokenizers trained on diverse languages. These tokenizers include three African tokenizers such as Afro-XLM tokenizer ($T_{Afro-xlm}$) [36], AfriBERTa tokenizer ($T_{AfriBERTA}$) [37], and Serengeti tokenizer ($T_{Serengeti}$) [38], as well as two well-known multilingual tokenizers such as xLM-R tokenizer (T_{XLM-R}) [4] and mBERT tokenizer (T_{MBERT}) [1]. In the rest of this paper, we denote the three African multilingual tokenizers as 'African tokenizers', and the other two multilingual tokenizers as 'general tokenizers'. The African tokenizers $T_{Afro-xlm}$, $T_{AfriBERTA}$, and $T_{Serengeti}$ are trained with 17, 11, and 517 African languages, respectively. The $T_{Afro-xlm}$ tokenizer also encompasses three widely spoken high-resource languages (HRLs) in Africa (e.g., English, French, and Arabic), and $T_{Serengeti}$ covers ten of the most widely spoken languages worldwide. The African tokenizers $T_{Afro-xlm}$, $T_{AfriBERTA}$, and $T_{Serengeti}$ have the vocabulary size of 250,002, 70,006, and 250,004, respectively. The general tokenizer T_{XLM-R} is trained with 100 languages including 8 African languages. Another general tokenizer T_{MBERT} is trained with 104 languages including 4 African languages. The general tokenizers T_{XLM-R} and T_{MBERT} have the vocabulary size of 250,002 and 110,350, respectively. Our primary objective in this experiment is to understand how well our language-specific tokenizers ($T_{lang-specific}$) perform compared to these multilingual tokenizers, and to examine the performance gap between African tokenizers and general tokenizers.

Table 5 presents a comparative analysis of sentiment classification performance for Swahili, Hausa, and Yoruba

TABLE 5. Sentiment classification accuracy results of African multilingual tokenizers (e.g., $T_{Afro-xlm}$, $T_{AfriBERTA}$, and $T_{Serengeti}$) and general multilingual tokenizers (e.g., T_{XLM-R} and T_{MBERT}). $T_{lang_specific}$ indicates the language-specific tokenizer.

Language	Classifier	$T_{lang-specific}$	$T_{Afro-xlm}$	$T_{AfriBERTa}$	$T_{Serengeti}$	T_{XLM-R}	T_{MBERT}
Swahili	SVM	85.51±0.01	84.61±0.01	86.81±0.01	**87.13±0.01**	86.40±0.01	83.26±0.01
	RF	90.58±0.01	90.68±0.01	90.84±0.01	**90.86±0.01**	90.79±0.01	**90.86±0.01**
	LR	89.80±0.01	83.02±0.01	89.12±0.01	**90.22±0.01**	86.98±0.01	84.32±0.01
	Avarage	88.30±0.01	86.77±0.01	88.59±0.01	**89.40±0.01**	88.72±0.01	86.48±0.01
Hausa	SVM	69.99±0.01	70.79±0.01	**71.05±0.01**	70.42±0.01	70.25±0.01	70.17±0.01
	RF	72.31±0.01	71.92±0.01	72.46±0.01	**72.50±0.01**	71.83±0.01	71.94±0.01
	LR	74.10±0.01	73.85±0.01	**74.17±0.01**	73.90±0.01	73.15±0.01	72.40±0.01
	Avarage	72.13±0.01	72.52±0.01	**72.89±0.01**	72.27±0.01	71.74±0.01	71.17±0.01
Yoruba	SVM	**64.96±0.01**	64.66±0.01	63.06±0.01	62.74±0.01	63.02±0.01	63.02±0.01
	RF	63.73±0.01	63.17±0.01	63.34±0.01	**63.83±0.01**	63.32±0.01	62.68±0.01
	LR	**68.78±0.01**	67.10±0.01	66.68±0.01	66.87±0.01	66.52±0.01	65.74±0.01
	Avarage	**65.82±0.01**	65.31±0.01	64.69±0.01	64.48±0.01	64.29±0.01	63.81±0.01

TABLE 6. News classification accuracy results of African multilingual tokenizers (e.g., $T_{Afro-xlm}$, $T_{AfriBERTA}$, and $T_{Serengeti}$) and general multilingual tokenizers (e.g., T_{XLM-R} and T_{MBERT}). $T_{lang_specific}$ indicates the language-specific tokenizer.

Language	Classifier	$T_{lang-specific}$	$T_{Afro-xlm}$	$T_{AfriBERTa}$	$T_{Serengeti}$	T_{XLM-R}	T_{MBERT}
Swahili	SVM	**57.20±0.01**	54.56±0.02	54.46±0.02	55.57±0.01	52.72±0.01	54.51±0.02
	RF	**56.25±0.01**	54.46±0.01	52.98±0.01	55.36±0.02	54.51±0.02	53.25±0.01
	LR	59.00±0.01	55.62±0.01	58.58±0.01	**59.37±0.01**	55.94±0.01	54.56±0.02
	Avarage	**57.15±0.01**	54.88±0.01	55.34±0.01	56.77±0.01	54.39±0.01	54.77±0.01
Hausa	SVM	**76.54±0.02**	70.04±0.01	69.06±0.01	69.21±0.01	72.68±0.01	72.36±0.01
	RF	85.08±0.01	**85.31±0.01**	85.00±0.01	85.12±0.01	83.11±0.02	84.09±0.01
	LR	84.76±0.01	**85.28±0.02**	85.20±0.02	85.12±0.02	84.37±0.02	84.61±0.01
	Avarage	**82.13±0.02**	80.87±0.01	79.75±0.01	79.82±0.01	80.72±0.01	80.02±0.01
Yoruba	SVM	**76.95±0.02**	53.96±0.02	53.54±0.02	52.07±0.01	58.17±0.01	61.52±0.01
	RF	**85.24±0.03**	82.26±0.02	84.82±0.02	84.15±0.03	82.56±0.01	83.35±0.02
	LR	**86.34±0.02**	84.76±0.02	85.85±0.02	85.12±0.02	83.96±0.01	84.27±0.01
	Avarage	**82.84±0.02**	73.99±0.03	74.07±0.02	73.11±0.01	74.23±0.01	76.71±0.02

languages using diverse multilingual tokenizers and classification models (SVM, RF, LR). The main finding in this result is that African tokenizers worked well for African languages. For example, $T_{Serengeti}$ and $T_{AfriBERTa}$ gave the best performance for Swahili and Hausa languages, respectively. For the Yoruba language, $T_{Afro-xlm}$ achieved comparable performance to the Yoruba-specific tokenizer $T_{lang-specific}$. This result is consistent with a previous finding that language models of LRLs may benefit from knowledge of other languages [35]. However, we found that general tokenizers (e.g., T_{XLM-R} and T_{MBERT}) generally underperformed compared to African tokenizers. This implies that African tokenizers benefit from linguistic similarities between many African languages. Therefore, multilinguality of tokenizers is not always preferable, and languages should be carefully selected for developing multilingual tokenizers for LRLs.

The news classification results are summarized in Table 6. This table provides a comparative analysis between our language-specific tokenizers and multilingual tokenizers. The result of Table 6 differs from the sentiment classification result in Table 5. The biggest difference is that the language-specific tokenizers ($T_{lang-specific}$) are the best, and especially for the Yoruba language, the performance gap between the Yoruba-specific tokenizer and the others is 6∼9%. Even the African tokenizers lag significantly behind the Yoruba-specific tokenizer. Furthermore, in the sentiment classification result in Table 5, the Yoruba-specific tokenizer was the only tokenizer that achieved better performance than other multilingual tokenizers. This suggests that the Yoruba language has characteristics that differ from those of Swahili and Hausa. First, Yoruba has few loanwords, which limits the benefits of multilinguality. Second, Yoruba is linguistically complex due to its many verb variations and a complicated intonation system, which uses different special symbols to represent tonal variations. For example, " ẹ̀dá" (creature) differs in meaning from "ẹ́dá" (mould) and " ẹ̀dá" (character) solely based on the tonal markings. To summarize, the results in Table 6 indicate that the multilinguality of a tokenizer may not be preferable if the target African language has few loanwords and complex linguistic properties.

C. LIMITATIONS AND FUTURE WORK

While this study provides valuable insights into the tokenization challenges for African low-resource languages (LRLs), we acknowledge that the dataset size, particularly for the downstream tasks, is a limitation. The sentiment classification (SC) and news classification (NC) datasets

were relatively small, which may affect the generalizability of the results.

In future research, we plan to expand the dataset size by incorporating additional labeled data for both SC and NC tasks. This will allow for more comprehensive evaluations of the tokenizers' performance. Additionally, exploring other downstream tasks beyond SC and NC will further strengthen the contribution of the work and improve the robustness of the tokenizers across diverse NLP applications for African languages.

Another limitation of this study is the use of traditional machine learning models (SVM, LR, RF) for tokenized text classification. While these models provide strong baseline performance, they may not fully capture the complex linguistic structures of African low-resource languages. In future work, we plan to adopt deep learning models such as LSTM, CNN, and RNN, which are well-suited for capturing sequential dependencies and learning more complex patterns in text data. These models are likely to improve performance, especially with larger datasets and more complex tasks.

V. CONCLUSION

This paper presented language-specific tokenizers for three African languages: Swahili, Hausa, and Yoruba. Our findings demonstrate that language-specific tokenizers outperform other monolingual tokenizers and multilingual tokenizers. Specifically, we found that the language-specific tokenizers perform better than other monolingual tokenizers trained with high-resource languages (HRLs). We also found that African multilingual tokenizers outperform those trained with a broader range of languages. Among the three African languages, the Yoruba-specific tokenizer is far better than the multilingual tokenizers. We provide an explanation for this result from a linguistic point of view. All of these findings imply the importance of considering the unique morphological and phonological characteristics of African languages during tokenization.

Based on the experimental results in this paper, we suggest an approach to tokenization for African languages as follows. When analyzing documents or building predictive models for African languages, a recommended starting point to use African multilingual tokenizers (e.g. Serengeti tokenizer) as suggested in [35], because most LRLs benefit from the rich resources of other languages. If the target language is Yoruba or any language with similar linguistic properties, we recommend creating a language-specific tokenizer. This will provide significant performance gains in NLP tasks.

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